Original Article

A Study on Assessment of Functional Status and Its Correlates Among the Geriatric Population in Rural India

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Abstract
Background: Functional ability is a proxy for healthy aging and is related not only to mental and physical health but also to social well-being. Identifying the determinants of functionality among the geriatric population can help evolve appropriate plans at both domiciliary and facility level. **Objectives:** To determine the magnitude of poor functional status among the geriatric population and its correlates. **Methods:** A community-based cross-sectional study was conducted among 246 geriatric people in a rural area of West Bengal from August to December 2022; data were analyzed using SPSS (version 16.0). Logistic regression was performed to identify correlates of poor functional status (activities of daily living [ADL] and instrumental ADL [IADL]). **Results:** 32.5% and 59.3% of participants were dependent for basic ADL and IADL, respectively. Age >70 years, female gender, less than primary education, widowed/separated status, joint family, poorest economic percentile, depression, and multimorbidity were associated with increased dependency. On multivariate analysis, age >70 years and depression remained significant for ADL; marital status, education, and family type additionally remained significant for IADL. **Conclusion:** Community-based comprehensive geriatric health and disability assessment should be provided, enabling early detection and delay of disabling disease progression, and supporting the well-being and independence of the elderly.

Key Words: Elderly, Functional status, Geriatric population, Predictors, Rural area

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Ethics Committee Clearance: Obtained from the Institutional Ethics Committee
Source(s) of Support: Nil
Conflict of Interest: None declared
How to cite this article: Sarkar C, Roy S. A study on assessment of functional status and its correlates among the geriatric population in rural India. J JMN Med Sci. December 2025;1(1):9–16.

I. Introduction

Aging is not uniformly experienced among aging adults. Some elderly people lead their life with a sense of fulfillment and satisfaction in their old age, while other older adults may become frail and experience declining physical abilities due to the decline of their physical abilities and social significance.[1] The life expectancy has increased from 40 years in 1951-1967 years in 2015, a person has 25 years more to live than he would have 50 years back.[2] According to SRS 2015, the percentage of the geriatric population in the rural area is 8.3, and inthe urban area, it is 8.4.[3] Due to globalization, most of the families results in breaking up of traditional families and formation of nuclear families that results to make the elderly to live a life in their own without taking any support.

The loss of functional ability typically associated with aging is absolutely related to a person's chronological age. There is no such "typical" older person.[4] Healthy aging is more than just the absence of disease. For most of the geriatric population, the maintenance of functional ability has the highest importance.[4] Functional status is usually defined in terms of restrictions on the ability to function independently in terms of basic ADL (ability to perform tasks of daily living) and another type instrumental ADL (IADL).[5] Functional activity is the performance related to not only mental and physical health; but it may also determine the aspect of social well-being.[6]

Longevity actually results in chronic diseases which affect the functional ability that may compromise the ability to pursue the daily routine activities, forming a situation with a need for assistance. So reduction of the severe disability is one of the best key to holding down health and social costs as well as to improve the quality of life and maintain a healthy well- being.[4] In the developed world there are numerous studies on geriatric care in developed settings. However, evidence from rural India remains limited, highlighting the need for context-specific health-system planning and high-quality physical and mental health services. Detection of the determinants of functionality components among the geriatric population

will help to assist in evolving with appropriate plans at both domiciliary and facility level to eliminate their sufferings and disabilities and to maintain a proper livelihood.

With this background, an attempt was made to assessthe functional status and the factors associated with it, among the geriatric population in a rural area of Singur, West Bengal.

II. Materials and Methods

The study place was the service area of Rural Health Unit and Training Centre, Singur under Field practice area of All India Institute of Hygiene and Public Health (AIIH&PH, Kolkata comprising of 64 villages. It is located in Singur block of Chandannagore sub-division in Hooghly District, with a distance of 50 km from the state capital of Kolkata; well connected by train (on Howrah-Tarakeshwar railway line) and road (besides Durgapur expressway and NH2). The study area contains 64 villages from seven Gram Panchayats, namely, Singur-I, Nasibpur, Anandanagar, Bagdanga-Chhinamore, Boinchipota, and Baruipara-Paltagarh.

This is a community-based cross-sectional study conducted among the geriatric population (60 years and above) in the Rural service area of A I I H & P H , Rural Health Unit and Training Centre Singur, Kolkata, which consists of 64 villages, 15-cluster technique was implemented, i.e., 15 villages were to be selected from 64 villages by the first stage of the cluster sampling. 15 clusters were selected from 64 villages as per PPS method by cluster sampling. 246 participants were recruited from 6 clusters. Total of 64 villages from which 6 villages were randomly selected, during the period of 5 months from August to December 2022. Taking reference of a similar study, where is 20% prevalence (p) for impairment in activities of daily living (ADL), with an absolute precision (d) of 5% and 95% confidence level with formula = z2pq/d2 (z = standard normal deviate at 95% confidence interval [CI]; q

= 100 - p), the estimated sample size was 246.[7] If there was no response received or consent could not be obtained from the study participant in the sampled household of the study area, then the next household was considered for the data collection. Furthermore, if one-household had two or more geriatric population persons, then a simple random sampling was applied to select the study participant to be included. If only one geriatric personwas staying in the house, and no other family members were available, and he was independently able to answer the interview schedule, he was included or else next household was chosen.

The study participants were selected through probability proportional to size sampling from

a line listing of the geriatric population in each village was prepared with the help of Voter's list and Primary sampling units, i.e., the clusters, which were in this case villages, were selected through Probability proportional to Population size method or PPS method. 2 stage cluster sampling was done to select the study participants [Table 1]. The data were collected using a pre-designed, pre-tested, structured schedule for the following:

Outcome variable

Functional status was assessed in terms of:

ADL: Refers to people's daily self-care activities. There are six domains of function: bathing, dressing, toileting, transferring, continence and feeding, adapted from Katz index questionnaire.[8] Cronbach's alpha for the Bengali version was 0.91

IADL: Activities that are not necessary for fundamental functioning, but they let an individual live independently in a community, comprises six domains: food preparation (female)/ shopping (male), housekeeping, laundry, the mode of transportation, the responsibility of own medications of the individual and the ability to handle finances, adapted from Lawton IADL Scale.[9] Cronbach's alpha for the Bengali version was 0.87.

Independent variables

Sociodemographic characteristics : age, sex, religion, caste, literacy, marital status, number of having children, economic status, and type of family were considered. The elderly people who are females or who belong to lower social economic status, who are illiterates are more likely to be neglected in aspects such as optimal health care, nutrition, and healthcareutilization in most of the aspects; this leads to chronic illness which resultsin increased dependence for ADL.

Self-reported morbidity profile (chronic diseases or any illness for 3 months as self-reported by participants.)

Depression: assessed using questions used to elicit the pattern of depressive symptoms in a study by Tamakoshi and Ohno.[10] Cronbach's alpha for the Bengali versionwas 0.73.

Face and content validity of the instrument was actually checked. The tool was translated into the local language (Bengali) maintaining semantic equivalence and pretesting of the questionnaire was done among 30 geriatric people residing in a village of Singur,RHUTC of AIIH&PH. The study was conducted after the clearance from the ethical consideration. Every participant had given written informed consent, after explaining to them the purpose

of the study and after ensuring the absolute confidentiality.

Operational definitions

In this study the outcome variable (functional status) was assessed in terms of both ADL and IADL which consists of six domains each. Participants were scored YES/NO for dependence in each of the six functions, if Yes (i.e.) activity needs supervision, direction, personal assistance or total care, they were allocated 1 point in that domain and if No, they were allocated 0 point (no assistance). The score ranks adequacy of performance in e a c h o f the six functions. The consolidated score of 6 indicates the severe functional impairment, were fully dependent on others and the score 0 indicates full function.

Study participants were categorized into two groups : according to their highest level of functioning of the study participants as:

Dependent (scores of 1-6, who needs assistance in atleast one of the activities) and

Independent (score 0, no assistance).

Data analysis

Data were analyzed using the SPSS Inc. Released 2007.SPSS for Windows, Version

16.0. Chicago, Illinois, US. Appropriate descriptive statistics, univariate and multivariate logistic regression analysis was performed to assess Functional status (ADL and IADL) and elicit its predictors with 95% CI ($P < 0.05$).

III. Results

In this study the mean age of the study participants was 71.82 with the range of 60-102. About 42.3% were females. 17.1% were scheduled caste. 43.1% of the study participants were illiterates. 43.5% of the study participants were widow/widowers. 73.2% lived in the joint family. About 48.7% of the study participants belonged to class IV socioeconomic status. Mean per capita income was Rs. 1823 among the study participants.

Tables 2 and 3 depicts the pattern of dependency for basic ADL and IADL and its relationship with gender showing significant association between gender with ADL (χ^2 = 22.4; d.f. = 1; P $\leq$ 0.05) and IADL (χ^2 = 20.6; d.f. = 1; P $\leq$ 0.05). The prevalence of dependency (at least one of the activities) for ADL was 32.5% among which 19 (7.7%) participants were highly dependent, i.e., need assistance to pursue all the six components of basic daily ADL, among which most, 18 (94.8%) participants were females. About 59.3% were dependent among which 31.3% were highly dependent for IADL.

Univariate logistic regression showed people aged >70 years, female gender, less than primary level education, widowed/separated, who lived in joint family, poorest percentile of economic status, who were depressed and the study participants suffered from multimorbidity (more than two chronic diseases) had the increased odds of dependency for ADL. In multivariate logit regression, after adjustment with significant variables, age adjusted odds ratio (AOR) (CI) (95% CI) - 20.6 (8-53.1) and depression AOR (CI) - 13 (4.5-37) remained significant. In this test as the Hosmer-Lemeshow statistic (goodness of fit test) became nonsignificant, the model is fit to explain the factors determining dependency for ADL among the study population. These factors explain about 51.8% variation in dependency for ADL (Nagelkerke R2 = 0.518) [Table 4].

For IADL, bivariate analysis showed every variable considered were significant except economic status (poorest), whereas multivariable analysis showed people aged 70 years 5.2 (2.3-11.4), widowed/separated 2.3 (1.1-5.2), less than primary level education 3.4 (1.5-7.7), joint family 2.6 (1.1-5.9), depression 2.8 (1.2-6.4) were significant among the study participants. The model is fit (Hosmer-Lemeshow statistic: nonsignificant) andit explains 53.5% variation in dependency for IADL [Table 5].

Table 1: *Selection of study participants through Probability Proportional to Size (PPS) sampling from selected villages*

S. No.	Village	Total elderly pop. (n)	Selected by PPS (n)
1	Anandnagar	313	71
2	Balitipa	65	15
3	Baghdanga	165	38
4	Diara	183	42
5	Chinnamore	272	61
6	Paltagarh	86	19
Total		**1084**	**246**

Table 2: *Distribution of study participants according to their functional status (n=246)*

Activity	Male, n (%)	Female, n (%)	Total, n (%)
ADL items			
Toilet	1 (0.7)	21 (20.2)	22 (8.9)
Bathing	3 (2.1)	20 (19.2)	23 (9.3)
Feeding	14 (9.9)	27 (26.0)	41 (16.7)
Dressing	19 (13.4)	36 (34.6)	55 (22.4)
Continence	17 (12.0)	30 (28.8)	47 (19.1)
Transfer within home	29 (20.4)	51 (49.0)	80 (32.5)
IADL items			
Food preparation/shopping	42 (29.6)	54 (51.9)	96 (39.0)
Housekeeping	51 (35.9)	54 (51.9)	105 (42.7)
Laundry	51 (35.9)	57 (54.8)	108 (43.9)
Responsibility for own medication	35 (24.6)	51 (49.0)	86 (35.0)
Transport (moving out of home)	51 (35.9)	67 (64.4)	118 (48.0)
Ability to handle finance	55 (38.7)	79 (76.0)	134 (54.5)

ADL: Activities of daily living; IADL: Instrumental activities of daily living. Figures denote participants dependent (needing supervision, direction, personal assistance, or total care) for that activity.

Table 3: *Distribution of study participants by number of dependent activities (n=246)*

(a) Activities of daily living (ADL)

No. of activities	Male, n (%)	Female, n (%)	Total, n (%)
6 (highly dependent)	1 (5.2)	18 (94.8)	19 (7.7)
5	2 (50.0)	2 (50.0)	4 (1.6)
4	7 (53.8)	6 (46.2)	13 (5.3)
3	9 (52.9)	8 (47.1)	17 (6.9)
2	2 (50.0)	2 (50.0)	4 (1.6)
1	8 (34.7)	15 (65.3)	23 (9.3)
0 (independent)	113 (68.0)	53 (32.0)	166 (67.5)
Mean$\pm$SD, median (range): 4.91$\pm$1.9, 6 (0–6)			χ^2=22.4, df=1, P<0.05

(b) Instrumental activities of daily living (IADL)

No. of activities	Male, n (%)	Female, n (%)	Total, n (%)
6 (highly dependent)	26 (33.7)	51 (66.3)	77 (31.3)
5	11 (78.6)	3 (21.4)	14 (5.7)
4	9 (100)	0 (0.0)	9 (3.7)
3	8 (72.7)	3 (27.3)	11 (4.5)
2	1 (9.1)	10 (90.9)	11 (4.5)
1	12 (50.0)	12 (50.0)	24 (9.3)
0 (independent)	75 (75.0)	25 (25.0)	100 (40.7)
Mean$\pm$SD, median (range): 3.36$\pm$2.6, 5 (0–6)			χ^2=20.6, df=1, P<0.05

Table 3: *Multivariable predictors of functional dependency in activities of daily living (ADL) among the elderly (n=246)*

Predictor	Adjusted OR	95% CI
Age >70 years	20.6	8.0–53.1
Depression	13.0	4.5–37.0

The final ADL model had Nagelkerke's R^2=0.518 and a non-significant Hosmer–Lemeshow goodness-of-fit test.

Table 4: *Multivariable predictors of functional dependency in instrumental activities of daily living (IADL) among the elderly (n=246)*

Predictor	Adjusted OR	95% CI
Age >70 years	5.2	2.3–11.4
Widowed/separated	2.3	1.1–5.2
Education <primary level	3.4	1.5–7.7
Joint family	2.6	1.1–5.9
Depression	2.8	1.2–6.4

The final IADL model had Nagelkerke's R^2=0.535 and a non-significant Hosmer–Lemeshow goodness-of-fit test.

IV. Discussion

The present study was conducted to assess the functional status ADL and IADL, which is an index of physical, mental and social well-being of geriatric people. Most of the studies among the elderly concentrate on morbidity and nutritional status, the ability to pursue basic daily activities is of much importanance, yet has been neglected, and very few studies have been conducted in this particular domain or aspect.

Prevalence of dependency

In the present study, dependency (for at least one of the ADL activities) was 32.4%, similar to study in rural Bengaluru (32.4%), whereas studies in rural Haryana (17.6%) and coastal Karnataka (6.5%) showed much less prevalence, it may be due to different study area and instrument used.[11-13] In the present study, among the highly dependent (score 6), most of them were females, whereas in a study in Puducherry showed males were highly dependent.[14] The studies carried out in developed countries such as Malaysia (14.4%), the

USA (15%), and Japan (20%) showed lesser prevalence of ADL disability than in the present study because of their social security system, including institution based unpaid care.[15-17] In the present study, 31.3% were highly dependent (all six activities) for IADL, whereas a study in Shimla showed 21.8% as in our country elderly people handover all the tasks, especially monetary issues, housekeeping totheir sons/daughter.[18] As most of the geriatric people become physically weak, even to move outside the home, they seek help. In the present study, IADL dependency(at least one of the IADL activities component) was 59.3%, whereas studies done in Poland and Norway showed the much less prevalence of at least one problem with IADL reported to be 43.19% and 19.9%, respectively.[19,20] In a study conducted in Spain showed 34.6% were categorized as dependent for at least one ADL and 53.5% for IADL which is similar to the present study, whereas a study conducted on oldest-old Brazilians showed less prevalence (28.7%) poor functional status which is due to the variation in measurement tool offunctional status.[21,22]

Predictors of poor functional status

In the present study and in other studies reviewed, ithas been observed that geriatric people or older age and female gender are important predictors of dependency. Aging leads to physical, mental and social weakening, as a result, they were prone to develop chronic illnesses which in turn lead to dependency and hence, the preventive methods should be provided at an early age, to have a healthy aging.[7,12,13,16-20] Females are still neglected in terms of food and care with a very little focus on their health and well-being, it is due to the gender-segregation related activities in our society where females were restricted to the kitchen work and they are not allowed to get involved in the economic reasoning and decisions. It has been observed that widow/separated were prone to have more dependency in this study and this was also seen in a study in state of Haryana.[12]. Another longitudinal study in India showed household wealth, religion and caste showed no clear association with ADL limitations, similar to this present study.[7]

Poor functional status is significantly associated with the condition of depression and chronic illnesses. In a study done in Malaysia showed depression (odds ratio [OR] - 1.8) and the presence of >2 chronic illnesses (OR - 8.4), whereas in the present study, there were higher odds of 12 and 6.6, respectively, as significant predictors of debilitating functional status.[15] Studies done in Brazil and Poland showed that depressed and multi-morbid persons having higher odds (OR - 2-3) of d e v e l o p i n g poor functional status.[20,21] In a longitudinal study done in America showed the development of high

depressive symptoms predicted with the greater risk of functional disability, explained 58.2% of the variation in risk factors, which was broadly comparable with the present study.[24] Chronic illness impairs the functional status and reduce physical activity, in turn, leads which may further reduce physical activity and contribute to a cycle of disability. In this study, older age and depression became to be imperative predictors after adjusting with all other variables of ADL. In addition to these, literacy and marital status were the other significant factors affecting IADL.

Strength

Most similar studies were conducted in clinical settings, whereas the current community-based study used a two-stage cluster sampling technique to select participants from 64 villages.

Limitation

However, other factors may also contribute to poor functional status. Prospective studies may provide a clearer picture; additionally, self-reported measures may have introduced subjective bias.

V. Conclusion

The study concluded that 32.5% and 59.3% were dependent for basic ADL and IADL respectively with geriatric people and depression the important predictorsof dependency for ADL. The functional disability among the elderly has become an important public health problem, has provided the impact t h a t they have it on their quality of life of individual, the family, and the health services. More focused emphasis should be given on the integration of geriatric care at the primary care level along with the other levels. There should be a provision for community-based comprehensive geriatric health assessment, as it enables geriatric people to avert illness at early stage, to delay the onset of disabling diseases and conditions and to provide basic domiciliary care and rehabilitation and specialist services at facility level. Social security measures like old age pension should be arranged more liberally and with a great aim for the aged people, especially for the disabled. This will ensure their active participation in the upliftment of the society and they thus will be less dependent on their progeny and will lead a healthy life.

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